

SKILLS

- **LLMs & RAG:** Transformers, Hugging Face, DeBERTa-v3, LangChain, LangGraph, sentence-transformers, Prompt Engineering
- **Vector Search:** ChromaDB, FAISS, Embeddings, Hybrid Retrieval
- **ML:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, SHAP, Pandas, NumPy
- **Deployment & MLOps:** FastAPI, REST APIs, Docker, LangSmith, LLMOps, Streamlit, Git, Hugging Face, SSE
- **Languages:** Python, C++, SQL, JavaScript
- **Soft Skills:** Cross-functional Collaboration, Technical Communication, Ownership, Fast Iteration, Adaptability

INTERNSHIP

· Vanillakart | [certificate](#)

Nov 2025 – Jan 2026

WordPress Developer

- Shipped and maintained production client websites on WordPress, owning responsive front-end development (HTML, CSS, JavaScript) end-to-end through deployment.
- Executed full content and asset migrations following SEO best practices, preserving all existing rankings with zero broken links post-launch.

PROJECTS

· C4Future — AI Carbon Footprint Predictor & Sustainability Advisor | [github](#) | [live demo](#)

Jun 2026

- Developed an end-to-end carbon footprint prediction platform using **XGBoost, SHAP, and Conformal Prediction**, achieving **82.6% pairwise ranking accuracy**, **0.82 Spearman correlation**, and **89.7% calibrated coverage** on 524 unseen products.
- Built a **RAG-powered sustainability advisor** (LangChain, ChromaDB, sentence-transformers, GPT-4o-mini) delivering citation-grounded recommendations from LCA and IPCC sources with cross-encoder reranking.
- Shipped a production **Django + Docker** app with explainable AI, real-time CO₂ estimation, and ~80 ms end-to-end prediction latency.

Tech: Python, LangChain, ChromaDB, Django, FastAPI, Docker

· AI Research Assistant — Autonomous Multi-Agent RAG Pipeline | [github](#) | [live demo](#)

Jun 2026

- Architected a 5-agent autonomous research system on a **LangGraph StateGraph** (Planner, Searcher, Reader, Critic, Writer) with conditional routing and a self-critique retry loop that re-searches coverage gaps before generating a sourced report.
- Engineered dual-layer memory — typed short-term graph state plus a long-term **ChromaDB** vector store with sentence-transformer embeddings — so each run retrieves semantically similar prior research and improves across sessions.
- Instrumented every step with **LangSmith** tracing; hardened the run with graceful degradation (JSON retries, failed-URL skips, memory-outage fallbacks); streamed live execution via **FastAPI + Server-Sent Events**.

Tech: Python, LangGraph, FastAPI, ChromaDB, BeautifulSoup, LangSmith

· Legal Clause Conflict Resolver | [github](#) | [live demo](#)

May 2026

- Fine-tuned **DeBERTa-v3-large** on NLI data for contradiction detection in legal and financial contracts, achieving **>85% validation accuracy** with ~15% fewer false positives vs. a zero-shot baseline.
- Built a retrieval layer with **FAISS + sentence-transformers** to surface candidate clause pairs, and implemented explainable AI using **cross-attention heatmaps** for interpretable legal evidence extraction.
- Deployed a **FastAPI** backend and analytics dashboard for real-time conflict analysis.

Tech: Python, PyTorch, Hugging Face Transformers, FAISS, sentence-transformers, FastAPI

ACHIEVEMENTS

- Contributed to **OpenCV** (open-source computer vision library) and authored an accepted documentation fix for **LlamaIndex** correcting a contradictory default model reference and a grammar error.

CERTIFICATES

· IBM – Python for Data Science and AI | [certificate](#)

June 2025

· IBM – ChatGPT-4 Prompt Engineering | [certificate](#)

June 2025

· Udemy – Master Generative AI & Generative AI Tools | [certificate](#)

June 2025

EDUCATION

Lovely Professional University

Punjab, India

Bachelor of Technology – Computer Science and Engineering; **CGPA: 8.08**

Hindi Higher Secondary School

Tripura, India

Intermediate – Percentage: 88% | Aug 2022

Holy Cross School

Tripura, India

Matriculation – Percentage: 94% | Aug 2020